Decision Errors

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MATH 240

Week 11, Lecture #2 — Ch. 14

Objectives

- **Identify** type I and II errors in a hypothesis test

Application of z-scores

Suppose I take a sample of 40 students whose SAT scores averaged 1200. Is this group of students significantly different from the population at $\alpha = 0.05$

$$\bar{X} \sim Normal(\mu = 1020, \sigma = 194^2/40 = 940.9)$$

$$\begin{aligned}
\text{p-value} &= 2 \times P(X > 1200) \\
&= 2 \times P\left(Z > \frac{1200 - 1020}{\sqrt{940.9}}\right) \\
&= 2 \times P\left(Z > 5.868144\right) \\
&= 2 \times \left(1 - P\left(Z \leq 5.868144\right)\right) \\
&= 4.4070 \times 10^{-9}
\end{aligned}$$

Computational vs. Mathematical Models

So far, we've seen the following models for sampling distribution:

- Computational: randomization, bootstrapping
  - Pros: no assumptions on data
  - Cons: computationally expensive; not always clear how to get these
- Mathematical: normal distribution
  - Pros: cleaner process and probabilities
  - Cons: need to verify assumptions (e.g. independence, sufficient sample size)

Activity: Concluding a Hypothesis Test

Suppose after conducting a hypothesis test on a difference of two proportions, you found that the p-value was 0.04. With a significance level of 0.05, you rejected the null hypothesis. Think about the following:

- Suppose there was no difference between the two proportions in reality.
  - What was the chance of this happening?
  - What could we have changed in our hypothesis testing procedures to avoid this situation from happening?
- Suppose we failed to reject the null hypothesis instead with a p-value of 0.04.
  - Would we feel comfortable failing to reject the null hypothesis with a p-value that low?
  - Say that there was a difference in means in reality. How could we possibly find the chance of this happening?

Consequences of Making a Decision

- Hypothesis testing allows us to make the best conclusion about a statistical scenario
- Unfortunately, we can never be perfect!
  - Have to introduce some error we are okay with making (i.e. significance level)
  - Discernibility level α represents the error of making the incorrect rejection when the null circumstance is actually true
    - ★ Also called the **type I error** (α)
    - ★ Probability of rejecting null hypothesis given the null hypothesis is true
- **Type II error** (β): probability of failing to reject null hypothesis given the alternative hypothesis is true

Decision Matrix

		Truth	
		H_0 true	H_0 is false
Decision	reject H_0	type I error	correct
	fail to reject H_0	correct	type II error

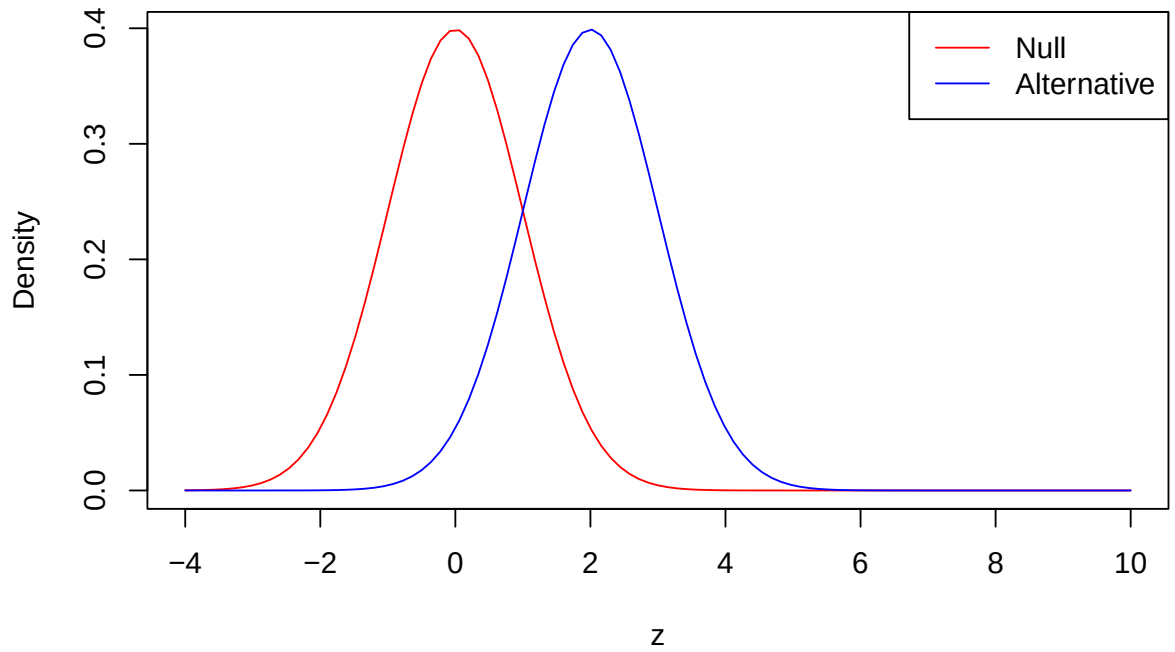
Type I and II Errors

$$\text{Type I error} = P(\text{rejecting } H_0 \mid H_0 \text{ true}) = \alpha$$

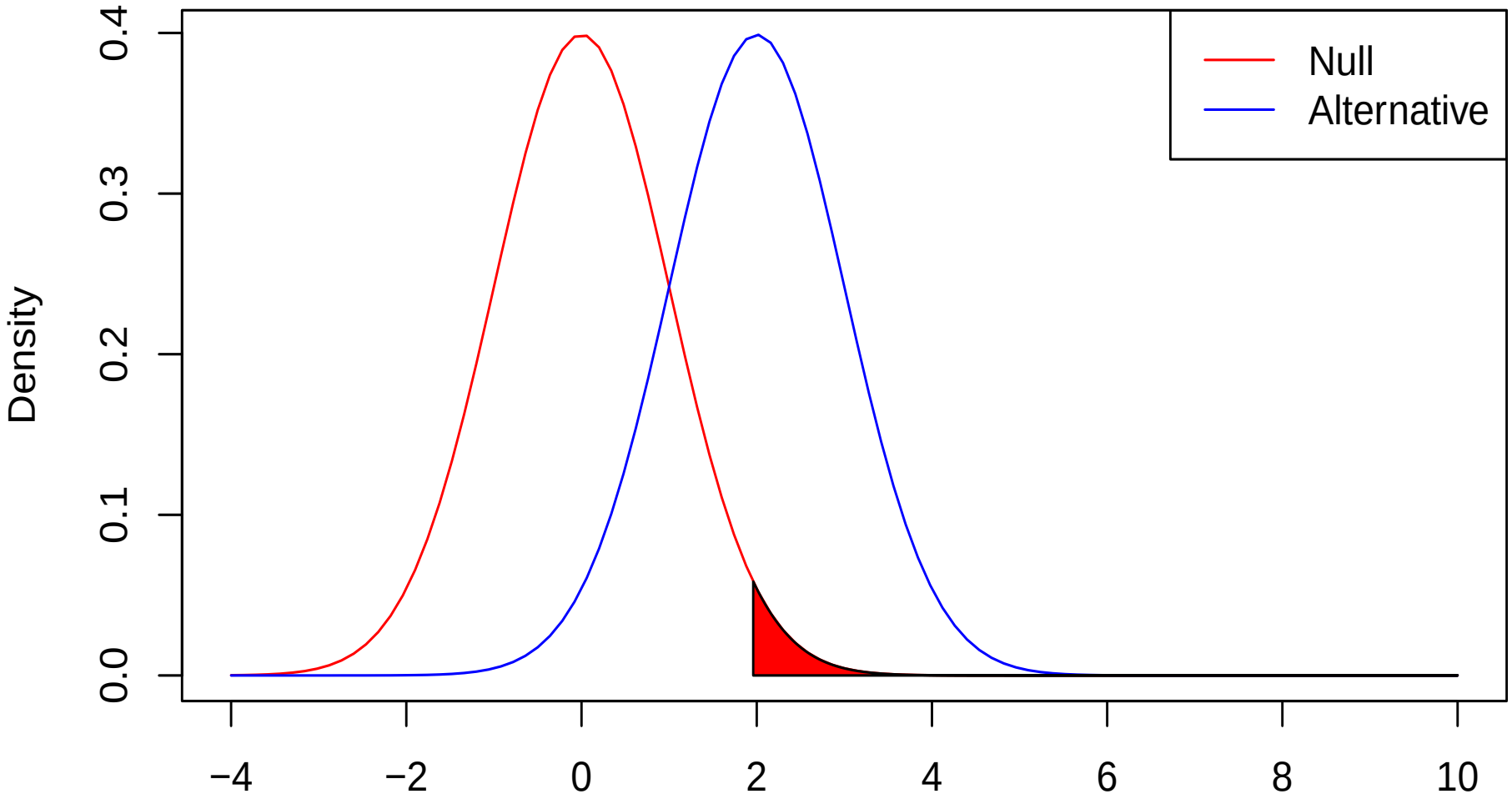
$$\text{Type II error} = P(\text{failing to reject } H_0 \mid H_1 \text{ true}) = \beta$$

Example

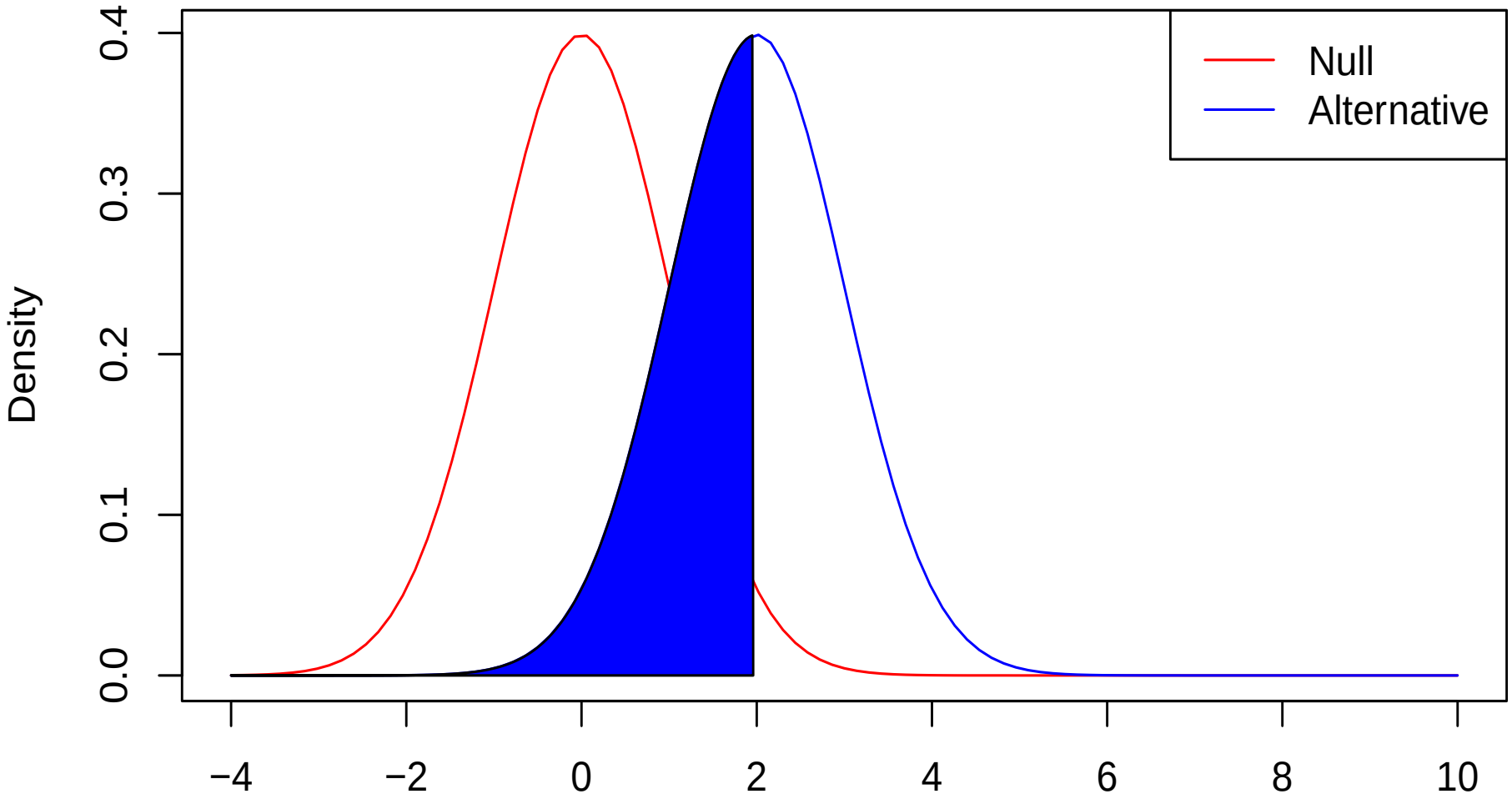
- We need to know both the null and alternative distributions
- Situation:
  - **Null:** Normal distribution centered at 0 with standard deviation 1
  - **Alternative:** Normal distribution centered at 2 with standard deviation 1
  - Say we have a significance level of 0.05



Visual of Type I Error



Visual of Type II Error



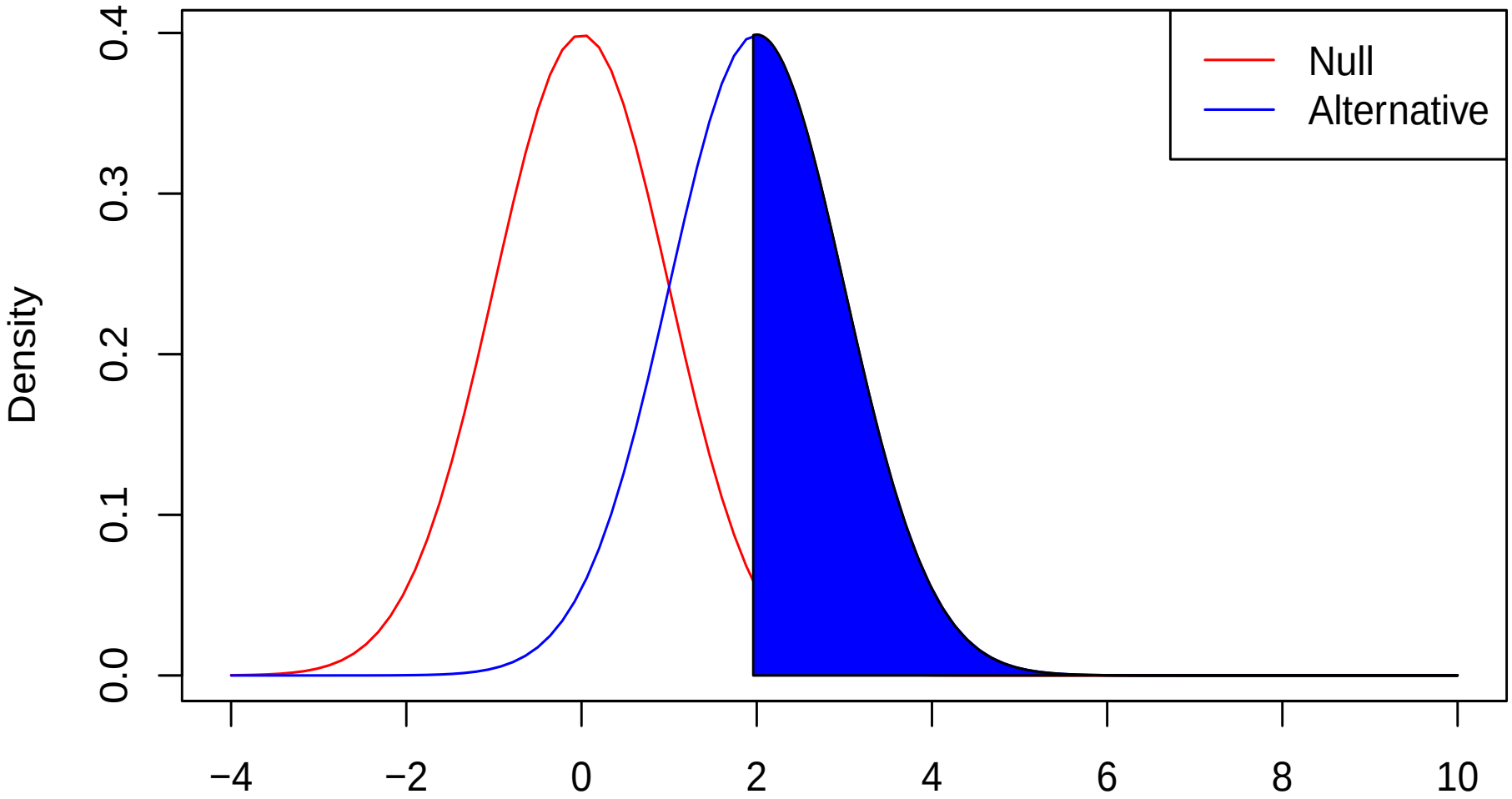
Power

- Useful in **experiment planning**
  - Collect enough data to detect important effects, yet reduce cost and risk
- Want to find out how likely we are to detect an effect we care about
  - In other words: we want high probability of rejecting the null hypothesis when the alternative distribution holds

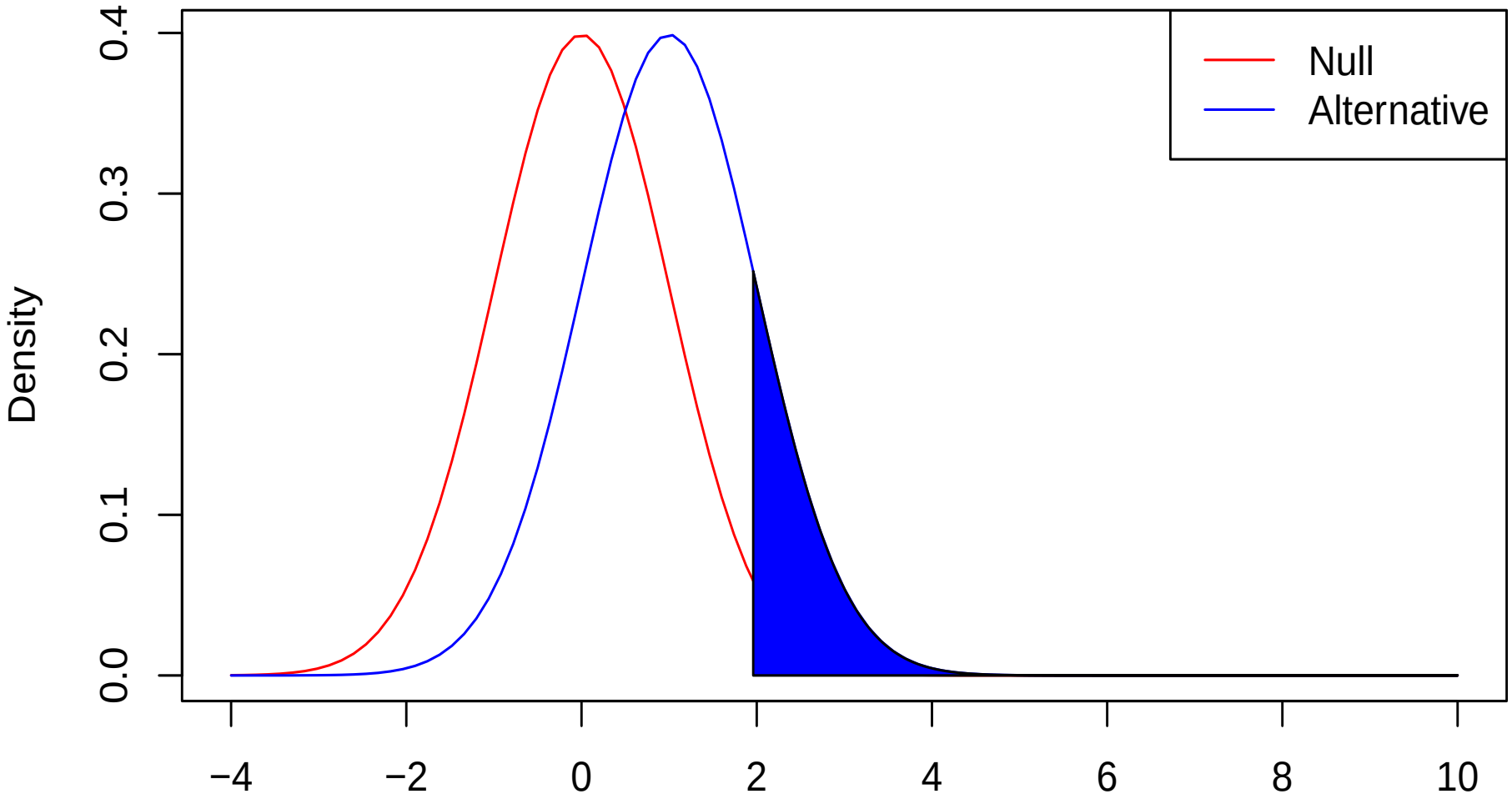
$$\begin{aligned}\text{Power} &= P(\text{rejecting } H_0 | H_1 \text{ true}) \\ &= 1 - P(\text{failing to reject } H_0 | H_1 \text{ true}) \\ &= 1 - \text{Type II error} \\ &= 1 - \beta\end{aligned}$$

- Affected by sample size, null/alternative means, error rate.

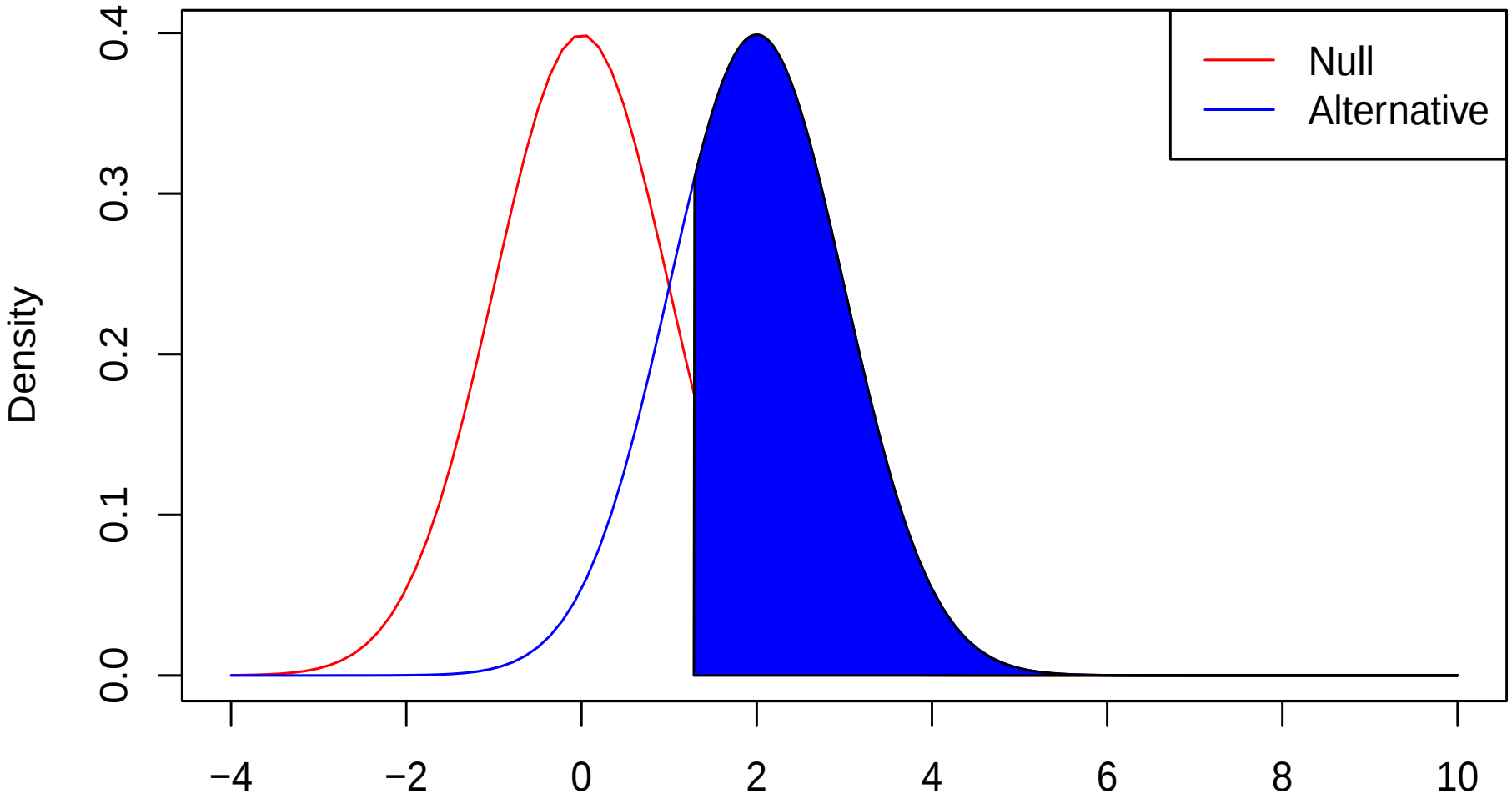
Example of Power (Visual)



Example of Power, with Means (Visual)



Example of Power, with Different α (Visual)



Example of Power, Sample Sizes (Visual)

